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Iodine Concentration in Milk, Ricotta Cheese, and Yogurt, and Their Contribution to Dietary Iodine Intake

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Abstract

Iodine deficiency is the leading preventable cause of neurological damage worldwide. Dairy foods represent an important dietary iodine source. This study aimed to assess iodine concentration in milk, ricotta cheese, and yogurt, and to evaluate their contribution toward the recommended daily iodine intake. Whole pasteurized milk (WM; n = 12), partially skimmed pasteurized milk (PM; n = 21), skimmed pasteurized milk (SM; n = 7), ricotta cheese (RC; n = 26), whole yogurt (WY; n = 13), and low-fat yogurt (LY; n = 15) were purchased in local stores. Samples were analyzed through inductively coupled plasma mass spectrometry for iodine quantification. After removing outliers, the final dataset comprised 11 WM, 19 PM, 7 SM, 26 RC, 13 WY and 15 LY samples. Data were investigated through a mixed model with iodine concentration as the dependent variable, product type as fixed effect, and brand as random effect. Low-fat yogurt exhibited the greatest estimated iodine concentration (293.76 µg/kg), while SM and WM exhibited the lowest (211.92 and 197.63 µg/kg, respectively). Based on these results, a serving of milk (250 g) would provide 31.82–39.08% of the average daily iodine requirement, a serving of ricotta (125 g) 21.66%, and a yogurt jar (125 g) 21.54–24.11%. These findings confirm the nutritional relevance of dairy products as primary iodine sources.

Keywords: dairy; nutrition; mineral; element

1. Introduction

Iodine is an essential microelement in human health, with particular regard to thyroid physiological function [1], thyroid hormone synthesis [2], and perinatal neurological development [3]. Based on international guidelines, the recommended daily iodine intake in healthy adults has been set at 150 µg/day [4]. However, this requirement varies considerably depending on sex, age, and physiological status, ranging from 90 µg/day in preschool children to 250 µg/day in pregnant and lactating women [4]. Failure to meet these thresholds may result in a wide range of health side effects, collectively referred to as iodine deficiency disorders. The magnitude and duration of iodine deficiency influence the extent of the associated disorders. Severe iodine deficiency during fetal life leads to increased risks of cretinism and infant mortality, while the same condition in adult life causes hypothyroidism and goiter [5]. Also, moderate iodine deficiency can have negative consequences on human health; indeed, women experiencing mild or moderate iodine deficiency during pregnancy are more likely to give birth to children with learning disabilities and lower verbal intelligence quotient [6].

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Even if it seems that iodine can be partially absorbed through inhalation [7,8], it has been well-established that the primary route of absorption is dietary. In addition to iodized salt, fish and seafood, there is widespread evidence that milk and dairy products are among the main sources of iodine in the human diet [9,10]. It is worth noting that knowledge and awareness about milk and dairy products as a source of iodine are still very limited among consumers and the general public [11]. Nonetheless, many scientific studies have focused on the characterization of milk iodine content. Niero et al. [12] quantified iodine in raw milk from different dairy species, including cow, buffalo, sheep, goat and donkey; Stevenson et al. [13] addressed the effect of technological treatments, farming systems, and season on milk iodine content; Costa et al. [14] investigated the genetic heritability of iodine content in bovine milk. Overall, it is widely acknowledged that milk iodine concentration is positively associated with the amount of iodine provided to animals through feed mineral supplementation [15,16]. Another important factor influencing milk iodine concentration is the use of iodized solutions for teat disinfection before and after milking, commonly referred to as "pre-dipping" and "post-dipping" [15]. Farming practices related to herd management (e.g., pasture versus confinement, or conventional versus organic) and environmental factors (e.g., seasonality) have been identified as additional sources of variation, influencing milk iodine concentration [16]. Regarding dairy products other than milk, van der Reijden et al. [17] observed that cheese iodine concentration is directly associated with milk iodine levels (i.e., the higher the iodine content in milk, the higher in cheese), while ripening seems to have negligible effect. Other authors have observed that iodized brine salting of soft, semi-hard, and hard cheeses is an effective strategy to increase iodine concentration in cheese [18]. Still, limited information is available regarding dairy products other than milk and cheese, such as ricotta cheese and yogurt, thus limiting knowledge on their contribution to the overall recommended daily iodine requirement. To the best of the author's knowledge, only Nerhus et al. [19] have attempted to characterize the iodine content of Norwegian whey cheese, although their study was limited to just nine samples. It is important to acknowledge that "ricotta cheese" is a well-defined dairy product, obtained by heat- and acid-induced coagulation of whey protein. In contrast, "whey cheese" refers to a broader category of dairy products primarily derived from whey, encompassing products with different processing conditions and physicochemical characteristics. Nerhus et al. [19] and van der Reijden et al. [17] have studied iodine concentration in yogurt matrix; nevertheless, Nerhus et al. [19] analyzed only six samples, while van der Reijden et al. [17] examined just four samples, again reflecting limited sample sizes.

Ricotta cheese and yogurt are widely consumed dairy products in many regions and contribute substantially to daily dairy intake. Including them in the estimation of dietary iodine intake is therefore important, as their consumption patterns may meaningfully affect total iodine intake, in addition to milk and other cheeses. Against this background, this study aims to (i) characterize iodine concentration and its variability across different dairy matrices, using milk as a benchmark alongside ricotta cheese and yogurt; (ii) investigate differences in iodine concentration among these dairy products; and (iii) evaluate their contribution in reaching the recommended daily iodine intake.

2. Materials and Methods

2.1. Sample Collection

Samples were collected based on market availability at the time of sampling, with the aim of maximizing representativeness and variability in terms of the number of brands and production batches within each dairy matrix. Whole pasteurized milk (WM; n = 12), partially skimmed pasteurized milk (PM; n = 21), skimmed pasteurized milk (SM; n = 7), ricotta cheese (RC; n = 27), whole yogurt (WY; n = 13), and low-fat yogurt (LY; n = 15) were

purchased from local stores (Veneto Region, Italy), totalling 95 samples and representing 95 production batches and 60 different commercial brands. All samples were labeled as conventional (i.e., no organic products were included in the sampling scheme). Although some commercial brands may be represented in more than one dairy matrix, the experimental design did not include systematic replication of the same brand across different product types. Therefore, the brand effect should be interpreted as the brand within the product type, accounting for variability among brands within each matrix rather than across matrices.

Based on nutritional labels, fat content ranged from 3.5 to 3.7% in WM samples, 1.5 to 1.8% in PM, 0.4 to 0.5% in SM, 8 to 12% in RC, 3 to 4% in WY, and 0.2 to 0.9% in LY. After purchasing, samples were transported at 4 °C to the Eurolab Laboratory (Bassano del Grappa, Italy), where iodine extraction and quantification were performed within 24 h.

2.2. Iodine Extraction and Quantification

Iodine extraction was carried out in the Eurolab laboratory (Bassano del Grappa, Italy), following the protocol validated by Niero et al. [12]. Samples of milk, ricotta cheese, and yogurt were diluted in 0.6% ultrapure ammonia solution (1:24) and incubated in a water bath at 90 °C for 1 h. After cooling at room temperature, samples were filtered using 0.45 µm syringe filters. Therefore, 5 mL of the filtered solution were further diluted in 0.6% ultrapure ammonia solution (1:1). The resulting solution was 50-fold diluted compared with the starting samples; this kept the expected sample salinity below 0.2%, as recommended for inductively coupled plasma mass spectrometry trials [20].

Iodine quantification was carried out using an inductively coupled plasma mass spectrometer (Ametek, Kleve, Germany) in the Eurolab laboratory (Bassano del Grappa, Italy), according to the analytical settings described by Niero et al. [12]. Method repeatability, calculated as the relative standard deviation of five consecutive measurements, ranged from 0.72 to 1.84% [12]. Method reproducibility, calculated as the relative standard deviation of 45 measurements performed on different days by different operators, was 4.01% [12]. The limit of detection and the limit of quantification of the analytical method were 5 and 15 µg of iodine per kg of milk, respectively [12]. Recoveries calculated on certified skim milk within a single day of analysis ranged from 87.41% to 98.90%, whereas the overall recovery, calculated as the mean across three days of analyses, was 92.55% [12].

2.3. Data Editing and Statistical Analysis

Preliminary data exploration showed that, within each type of matrix, iodine concentration was normally distributed. Outliers for iodine concentration were defined as values exceeding three standard deviations from the mean. Based on this criterion, one outlier was found for WM, two for PM, and one for RC, while no outliers were found for SM, WY, and LY. Outliers were set as missing values and were not considered for the subsequent statistical analyses. Therefore, the final dataset comprised 11 WM, 19 PM, 7 SM, 26 RC, 13 WY, and 15 LY samples.

Descriptive statistics of iodine concentration in WM, PM, SM, RC, WY, and LY were calculated through the PROC MEANS of SAS software version 9.4 (SAS institute Inc., Cary, NC, USA). Sources of variation for iodine concentration were investigated according to the following linear mixed model, implemented through the PROC MIXED of SAS software:

$$Y_{ijk} = \mu + M_i + B_j + e_k$$

where Y_{ijk} is the dependent variable (i.e., iodine concentration); μ is the overall intercept of the model; M_i is the fixed effect of the ith type of matrix (i = WM, PM, SM, RC, WY, and LY); B_j is the random effect of the jth brand (j = 1–60); e_k is the random residual ~N(0, σ^2_e),

where σ^2_e is the residual variance. Differences between least squares means of iodine concentration were assessed using post hoc multiple comparison ($p < 0.05$).

Coverage of the recommended daily iodine intake provided by each sample of WM, PM, SM, RC, WY, and LY was calculated as the percentage ratio between the iodine content of a typical serving (250 g for milk, 125 g for ricotta cheese, and 125 g for yogurt) [21], and the recommended daily iodine intake for adults (150 µg). Descriptive statistics for coverage of recommended daily iodine intake provided by a standard serving of WM, PM, SM, RC, WY, and LY were calculated through the PROC MEANS of SAS software version 9.4.

3. Results and Discussion

3.1. Descriptive Statistics of Iodine Concentration

Descriptive statistics of iodine concentration in WM, PM, SM, RC, WY, and LY are reported in Table 1. Among studied matrices, LY exhibited the greatest average iodine concentration (289.33 µg/kg; Table 1) followed by RC and WY, having similar numerical values (260.00 and 258.46 µg/kg, respectively; Table 1). In general, the milk matrix showed lower average iodine concentrations compared to the other dairy matrices considered in this study, with PM, SM, and WM averaging 234.47, 207.86, and 190.91 µg of iodine per kg, respectively (Table 1). Average iodine concentrations observed in the present study for LY and WY are about two times greater than the values reported by Nerhus et al. [19] who studied the iodine content of different Norwegian dairy products. Also, van der Reijden et al. [17] reported lower iodine concentrations in yogurt while investigating the effects of feed iodine levels and milk processing on iodine concentrations in bovine milk and related dairy products. In any case, the high iodine concentrations observed in the present study for LY (up to 640 µg/kg) may reflect variation in the iodine content of the raw milk and differences in processing, such as the concentration of milk solids or addition of whey. Although the specific sources of this variability were not investigated in the present study, these results highlight the importance of future studies to elucidate the factors influencing iodine concentration in yogurt. Average iodine concentration observed in the present study for RC is hardly comparable with the existing literature, since very scarce information is available. To the best of the author's knowledge, only Nerhus et al. [19] have reported iodine concentration in Norwegian brown whey cheese, including three samples each of "Gudbrandsdalsost", "Fløtemysost", and "Ekte geitost". The iodine concentrations they reported are extremely high, being about five to seventeen times greater than those observed in the present study for RC, to the extent that regular consumption of such products over time could pose a potential risk to human health, as excessive iodine intake may lead to toxicity [16]. This substantial difference is most likely explained by the distinct technological processes involved in the production of RC and Norwegian brown whey cheese. Average iodine concentrations for WM, PM, and SM in the present study align with previously published data. In fact, iodine concentration in retail milk is not subject to standardization and is therefore highly variable, ranging from 91 µg/L [22] to 489 µg/kg [23]. Such great variability in retail milk has been associated with several factors, the most relevant of which include animal feed, season of production, and farming system [16]. Although a detailed analysis of factors affecting iodine variability falls outside the objectives of this study and cannot be undertaken without additional data (e.g., seasonality, feeding strategies, farming system, processing conditions), it is important to acknowledge that these elements still contribute to the differences observed across dairy matrices. In this view, their potential impact should be kept in mind when interpreting the present findings.

In the present study, non-milk matrices (i.e., LY, WY and RC) exhibited higher coefficients of variation (48.76, 37.15, and 36.12%, respectively; Table 1) compared with milk matrices (i.e., WM, PM, and SM). These findings suggest that LY, WY, and RC are less

standardized in terms of iodine concentration compared to milk products, likely arising from differences in manufacturing practices and technological processes. In particular, variation in fat content, fermentation conditions, formulation (e.g., milk standardization or whey-to-milk ratios), and the possible use of iodized salt during processing may contribute to the observed heterogeneity, as also suggested by the wide range of fat contents reported on product nutritional labels.

Table 1. Descriptive statistics of iodine concentration (µg/kg) in whole pasteurized milk (WM), partially skimmed pasteurized milk (PM), skimmed pasteurized milk (SM), ricotta cheese (RC), whole yogurt (WY), and low-fat yogurt (LY).

Matrix	N	Mean (µg/kg)	Standard Deviation	Coefficient of Variation (%)	Minimum	Maximum	95% Confidence Interval
WM	11	190.91	59.78	31.32	90.00	270.00	150.75–231.07
PM	19	234.47	65.91	28.11	115.00	335.00	202.70–266.24
SM	7	207.86	32.26	15.52	170.00	260.00	178.03–237.69
RC	26	260.00	93.90	36.12	100.00	540.00	222.07–297.93
WY	13	258.46	96.01	37.15	165.00	480.00	200.44–316.48
LY	15	289.33	141.09	48.76	155.00	640.00	211.20–367.47

3.2. Least Squares Means for Iodine Concentration

Least squares means for iodine concentration in WM, PM, SM, RC, WY, and LY are depicted in Figure 1. Post hoc comparisons performed on least squares means indicated that LY had the highest iodine concentration (293.76 µg/kg), which was significantly higher than SM and WM (211.92 and 197.63 µg/kg, respectively). Intermediate iodine concentrations were observed for WY (265.42 µg/kg), RC (251.52 µg/kg), and PM (237.44 µg/kg), which did not differ significantly from either the highest or the lowest values, reflecting their intermediate position among the matrices. Overall, the results of this study indicate that processed dairy products tend to have higher iodine concentrations than liquid milk, which is consistent with the conclusions drawn by van der Reijden et al. [17]. Indeed, iodine concentrations in processed dairy products increase as a result of the concomitant moisture loss occurring during manufacturing [17]. An additional insight from this analysis is that low-fat dairy products tend to have slightly higher iodine concentrations compared to their full-fat counterparts. This trend is observed both in LY and PM, which exhibit higher iodine concentrations than WY and WM, respectively (Figure 1). Such a negative association, where lower fat content corresponds to higher iodine concentration, has already been described in the literature [12,14]. Indeed, since iodine compounds are water-soluble, they tend to be more concentrated in low-fat products, which have a greater water fraction, compared to full-fat products, which have a smaller water fraction [23].

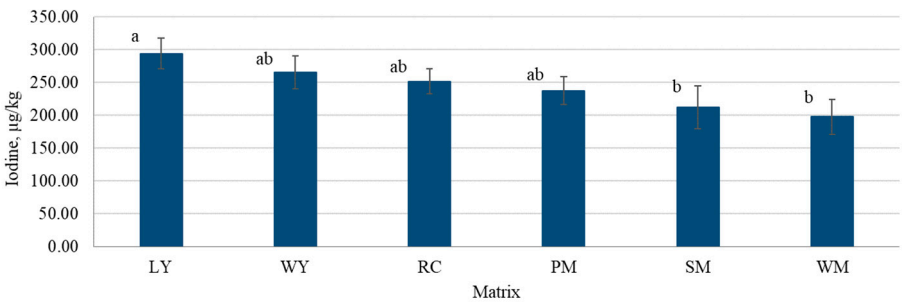

Figure 1. Least squares means (and standard errors) for iodine concentration in whole pasteurized milk (WM), partially skimmed pasteurized milk (PM), skimmed pasteurized milk (SM), ricotta cheese (RC), whole yogurt (WY) and low-fat yogurt (LY). Different letters indicate significantly different estimates ($p < 0.05$).

3.3. Descriptive Statistics of Iodine Intake from Milk, Ricotta Cheese, and Yogurt

Iodine concentration in milk, ricotta cheese, and yogurt represents only one side of the coin regarding their contribution to the overall recommended daily intake. Indeed, an important factor is the amount of these foods consumed in the diet, both in terms of serving size (e.g., on weight basis, a serving of milk is typically much larger than a serving of ricotta cheese) and consumption frequency (e.g., on a weekly or monthly basis, milk is often consumed daily, while ricotta cheese is not). In this light, descriptive statistics for the contribution to the recommended daily iodine intake provided by a standard theoretical serving of WM, PM, SM, RC, WY, and LY are reported in Table 2. In particular, it can be noticed how a typical serving of milk (250 g) provides between 31.82% and 39.08% of the average daily iodine requirement, for WM and PM, respectively. This contribution is systematically higher than that provided by a 125 g serving of RC (accounting for 21.66% of the average daily iodine requirement) and by a 125 g serving of WY and LY (accounting for 21.54 and 24.11% of the average iodine requirement, respectively). This divergence could become even more pronounced by considering that milk is usually consumed more frequently (once or more per day) compared to ricotta cheese and yogurt, although this pattern may vary depending on individual dietary habits. Interestingly, even when considering the maximum values for coverage of the recommended daily iodine intake, the percentages ranged from 55.83% for PM to 40.00% for WY (Table 2), still indicating a substantial yet balanced iodine supply, which is well below excessive doses of iodine intake. In conclusion, even though the results of the present study highlight that milk is characterized by a lower absolute iodine concentration compared to ricotta cheese and yogurt (Figure 1), it should be noted that when typical serving sizes are considered, milk still provides the most substantial contribution in terms of percentage of the recommended daily iodine intake (Table 2), reaffirming its predominant role in this context. In line with the conclusions reported by Niero et al. [16], these findings further support the central role of milk and dairy products in addressing sufficient iodine intake. Therefore, improving the awareness of the general public on dairy products as dietary iodine sources, along with better controlling factors influencing iodine concentration in dairy products, may contribute substantially in establishing effective policies, promoting adequate iodine intake [11].

Table 2. Descriptive statistics for coverage (%) of recommended daily iodine intake (150 µg/day) provided by a standard serving of whole pasteurized milk (WM; 250 g), partially skimmed pasteurized milk (PM; 250 g), skimmed pasteurized milk (SM; 250 g), ricotta cheese (RC; 125 g), whole yogurt (WY; 125 g), and low-fat yogurt (LY; 125 g).

Matrix	N	Mean (%)	Standard Deviation	Coefficient of Variation (%)	Minimum	Maximum	95% Confidence Interval
WM	11	31.82	9.96	31.32	15.00	45.00	25.12–38.51
PM	19	39.08	10.99	28.11	19.17	55.83	33.78–44.37
SM	7	34.64	5.38	15.52	28.33	43.33	29.67–39.61
RC	26	21.66	6.26	36.11	8.33	46.25	17.97–24.39
WY	13	21.54	8.00	37.15	13.75	40.00	16.70–26.37
LY	15	24.11	11.76	48.76	12.92	53.33	17.60–30.62

4. Conclusions

Overall, this study confirmed the importance of milk and dairy products as primary iodine sources in the daily diet. Among studied matrices, LY exhibited the greatest iodine concentration, SM and WM the lowest, while WY, RC, and PM were characterized by intermediate values. This picture changes when considering the contribution of each typical serving size to the recommended daily iodine intake, with milk covering the highest portion of daily iodine requirement, followed by RC and LY. The main limitations of this study include the absence of seasonal data, the lack of information on animal feed and

herd management, the focus on Italian products only, and the relatively small sample sizes within matrix. Future research should be conducted at the dairy plant level, encompassing the entire production chain (i.e., from raw milk to final dairy products, including whey and by-products), in order to understand how and to what extent iodine is transferred at each stage of industrial processing (e.g., thermal treatments, fermentation, and brining).

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Abbreviations

The following abbreviations are used in this manuscript:

WM	Whole pasteurized milk
PM	Partially skimmed pasteurized milk
SM	Skimmed pasteurized milk
RC	Ricotta cheese
WY	Whole yogurt
LY	Low-fat yogurt

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